

Treatment of Chronic Venous Insufficiency

Most experts agree on the essential role of compression in treating chronic venous insufficiency. Hence, unless there are contraindications such as arterial occlusive disease or abnormal ankle-brachial indexes, compression therapy should remain a core component of any treatment regimen for chronic venous disease.

Toward the end of the nineteenth century, Paul Gerson Unna demonstrated that applying a compression bandage to the distal extremity could heal venous ulcers in ambulatory patients using a zinc-impregnated gauze wrap—commonly known as the "Unna boot." While this method has been well-documented in the literature, achieving consistent graduated compression with the original Unna boot was challenging, and patient compliance was often poor. To address these limitations, a self-adherent second layer (e.g., Coban™) was added over the zinc paste layer, forming what is now referred to as the "Duke boot." This modification allows for easier application and delivers reproducible, graduated inelastic compression. Compression bandaging to counteract venous outflow due to perforator incompetence remains the cornerstone of venous ulcer management.

Dressings should be changed weekly, or more frequently if the wound is heavily exudative. New granulation tissue is sensitive to cytotoxic agents, so most topical antiseptics should be avoided. A useful principle is: *do not put anything into a wound that you would not put into your eye*. Gentle cleansing with tap water or saline is sufficient. Whenever possible, dressing changes should be performed with the patient in a supine position.

Mechanical Therapy

Mechanical therapy is the mainstay of treatment for all clinical stages of chronic venous insufficiency.

- Leg elevation
- Compression stockings
- Compression bandages
- Intermittent pneumatic compression pumps

Drug Therapy

Pharmaceutical treatments target specific aspects of chronic venous insufficiency:

- **Aspirin** (300–325 mg/day) may enhance venous ulcer healing
- **Pentoxifylline** has been shown to improve ulcer healing rates
- **Topical steroids** (for stasis dermatitis)
- **Horse chestnut seed extract** (standardized to 50 mg escin twice daily) – effective for short-term relief of leg pain and swelling
- **Diuretics** – may be used short-term for severe edema

Dressings

- Wet-to-dry non-adherent dressings
- Occlusive hydrocolloid or gel dressings
- Zinc paste-impregnated bandages (Unna boot)

Surgical Options

- Sclerotherapy
- Long saphenous vein stripping
- Saphenofemoral/saphenopopliteal junction disconnection
- Calf varicosity avulsions

- Subfascial endoscopic perforator veinligation
- Endovenous laser therapy
- Radiofrequency ablation

Infection Management

Although all wounds harbor microorganisms, routine use of systemic antibiotics does not improve outcomes in uncomplicated venous ulcers. If cellulitis is suspected, empiric antibiotic therapy with coverage for *Staphylococcus aureus* and *Streptococci* is indicated. Topical antibiotics, particularly mupirocin, are effective for bacterial folliculitis caused by these organisms. Given the rising prevalence of community-acquired methicillin-resistant *S. aureus* (CA-MRSA), culture and sensitivity testing are recommended for suspected skin and soft tissue infections.

Long-Term Management

Venous disease is progressive and irreversible. After ulcer healing, patients must be educated about the lifelong need for hemodynamic support. Graduated compression stockings providing 30–40 mmHg at the ankle should be custom-fitted and worn daily. It is important to avoid placing elastic stockings on acutely edematous or tender limbs, as this can impair circulation and cause discomfort.